Teacher Wu Kuan | See It at a Glance #9 — Why sin 30° = 1/2

English lecture notes · YouTube Short NTzCS0CHC-U

1. The trap question

Why is sin 30° exactly 1/2?

2. The method

• Equilateral triangle, side 2; cut down the middle.
• Angles 30°–60°–90°; sides 1, √3, 2.
• sin 30° = opposite/hypotenuse = 1/2.

3. Parent check method

Ask your child to draw the half equilateral triangle and read off all six values.

Practice

1. sin 30°
2. cos 30°
3. tan 30°
4. sin 60°
5. A right triangle has hypotenuse 10 and a 30° angle. Find the side opposite 30°.

Answers

1. 1/2
2. √3/2
3. √3/3
4. √3/2
5. 5

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